Lecture 18 (*April 2*) *Reading: Section 8.2*

Viewing: Lecture 31

In the previous Lecture, we always worked in the same basis on the domain (source) of the linear transformation ϕ as on the codomain (target). But now we will see what happens if we allow different bases on the domain and codomain. So let us consider a general linear transformation:

$$\phi : \mathbb{R}^n \to \mathbb{R}^m$$

which is represented by the $m \times n$ matrix $A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$ in the standard bases:

$$\phi(x_1\boldsymbol{e}_1 + \cdots + x_n\boldsymbol{e}_n) = (a_{11}x_1 + \cdots + a_{1n}x_n)\boldsymbol{e}_1 + \cdots + (a_{m1}x_1 + \cdots + a_{mn}x_n)\boldsymbol{e}_m \tag{144}$$

Since $\mathbb{R}^m$ and $\mathbb{R}^n$ are (in general) different vector spaces, it will often make sense to change the basis in the input $\mathbb{R}^n$ independently of the change of basis in the output $\mathbb{R}^m$. So let us fix bases:

$$\boldsymbol{v}_1, \dots, \boldsymbol{v}_n \text{ of } \mathbb{R}^n \qquad \text{and} \qquad \boldsymbol{w}_1, \dots, \boldsymbol{w}_m \text{ of } \mathbb{R}^m$$

If you wish to express the linear transformation ϕ in these new bases, consider the matrices:

$$V = \left[\ \boldsymbol{v}_1 \mid \dots \mid \boldsymbol{v}_n\ \right] \qquad \text{and} \qquad W = \left[\ \boldsymbol{w}_1 \mid \dots \mid \boldsymbol{w}_m\ \right]$$

As we have seen in the previous lecture, V is the change of basis matrix from the basis $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ to the basis $\boldsymbol{e}_1, \dots, \boldsymbol{e}_n$, and W is the change of basis matrix from the basis $\boldsymbol{w}_1, \dots, \boldsymbol{w}_m$ to the basis $\boldsymbol{e}_1, \dots, \boldsymbol{e}_m$. Therefore, W^{-1} is the change of basis matrix from the basis $\boldsymbol{e}_1, \dots, \boldsymbol{e}_m$ to the basis $\boldsymbol{w}_1, \dots, \boldsymbol{w}_m$. We conclude that the linear transformation ϕ is represented by the matrix:

$$\boxed{B = W^{-1}AV} \tag{145}$$

in the new bases $\boldsymbol{v}_1, \dots, \boldsymbol{v}_n$ and $\boldsymbol{w}_1, \dots, \boldsymbol{w}_m$. Explicitly, this means that if:

$$B = \begin{bmatrix} b_{11} & \dots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \dots & b_{mn} \end{bmatrix}$$

then formula (144) is equivalent to:

$$\phi(x_1\boldsymbol{v}_1 + \cdots + x_n\boldsymbol{v}_n) = (b_{11}x_1 + \cdots + b_{1n}x_n)\boldsymbol{w}_1 + \cdots + (b_{m1}x_1 + \cdots + b_{mn}x_n)\boldsymbol{w}_m \tag{146}$$

Formula (145) is called the **change of basis** formula.

The ability to change bases gives us a lot of freedom in studying matrices. Basically, you can always change bases in order to simplify any given matrix, and how far you go depends on the problem at hand. For example, in the square $n \times n$ case, <u>almost</u> any matrix is similar to a diagonal matrix:

$$A = V \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} V^{-1} \tag{147}$$

In future weeks, we will see the meaning of the numbers $d_1, \ldots, d_n$ (the eigenvalues of A) and of the columns of the matrix V (the eigenvectors of A). We will also learn how to eliminate the word "almost" in the previous paragraph, by studying Jordan normal forms.

Example 6. *Consider the linear transformation $\phi : \mathbb{R}^3 \to \mathbb{R}^2$ represented by the matrix:*

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix}$$

In the standard bases, this linear transformation sends a vector:

$$x_1 \boldsymbol{e}_1 + x_2 \boldsymbol{e}_2 + x_3 \boldsymbol{e}_3 \quad \textit{to} \quad (x_1 + x_2 + 2x_3)\boldsymbol{e}_1 + (2x_1 + 3x_2 + 4x_3)\boldsymbol{e}_2$$

But can we change bases so that ϕ is represented by a "nicer" matrix than A? Say we wanted to find bases $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ of $\mathbb{R}^3$ and $\boldsymbol{w}_1, \boldsymbol{w}_2$ of $\mathbb{R}^2$ such that the same linear transformation ϕ is represented by the simple matrix:

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

in the new bases. In other words, the same linear transformation ϕ should send a vector:

$$x_1 \boldsymbol{v}_1 + x_2 \boldsymbol{v}_2 + x_3 \boldsymbol{v}_3 \quad \textit{to} \quad x_1 \boldsymbol{w}_1 + x_2 \boldsymbol{w}_2$$

Fortunately, the change of basis formula (145) *tells us exactly what we need to do. We need to find invertible matrices V and W such that:*

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = W^{-1} \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix} V$$

or in other words:

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix} = W \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} V^{-1} \tag{148}$$

We don't need to just guess V and W, since we already have a system in place for simplifying a matrix A: Gauss-Jordan elimination. Running this algorithm on the original matrix A gives us:

$$E_{12}^{(-1)} E_{21}^{(-2)} \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix}$$

By moving the elimination matrices to the right, we get:

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix} = E_{21}^{(2)} E_{12}^{(1)} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}}_{\textit{call this matrix } W} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix}$$

The right-most matrix isn't quite as simple as B, and we cannot make it any simpler using row operations. But we can do column operations, which we know corresponds to multiplication on the right with elimination/diagonal/permutation matrices. In the case at hand, this is quite easy to do:

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix} E_{13}^{(-2)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

By moving the elimination matrix to the right, we get:

$$\begin{bmatrix}1 & 0 & 2\\0 & 1 & 0\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\end{bmatrix} E_{13}^{(2)} = \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\end{bmatrix} \underbrace{\begin{bmatrix}1 & 0 & 2\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}}_{\textit{call this matrix } V^{-1}}$$

Therefore, we have achieved the formula $A = WBV^{-1}$ *using the matrices:*

$$V = \begin{bmatrix}1 & 0 & -2\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix} \quad \left(\textit{this is equivalent to } V^{-1} = \begin{bmatrix}1 & 0 & 2\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}\right) \qquad \textit{and} \qquad W = \begin{bmatrix}1 & 1\\2 & 3\end{bmatrix}$$

Hence the desired bases have $\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3$ *and* $\boldsymbol{w}_1, \boldsymbol{w}_2$ *equal to the columns of* V *and* W*, respectively.*